Spring ’24 – User Research Methods – Professor Charles Raffaele
Observation Design Document – Darryl Hutchinson

INTRODUCTION & RESEARCH QUESTIONS

Though they have great potential for entertainment and learning, video games are typically very complex and it can be easy for inexperienced players to get overwhelmed and lose site of the objectives the designer wants them to experience. To better understand intuitive level design and which game elements do the best job of capturing player attention regardless of experience level such that they don’t get lost and can progress through an experience, I want to analyze the upcoming video game, Europa.

Designed and developed by Helder Pinto and Novadust studios, Europa is a bright and colorful Sci-Fi adventure game with a heavy influence of painterly Ghibli style visual animation, smooth freeform movement and traversal with low stress exploration. The game has been in development since 2017 and is currently available in demo form with a scheduled release window of spring 2024.

I am interested in exploring pacing/pathing and environmental storytelling. I would like to know specifically how game elements in the game world can communicate which ways players can travel without using text or unnatural UI markers. Since video games are relatively complex bundles of systems that are easy to get lost within, I want to try and understand how to better lead the player along an experience with natural elements that help maintain immersion.

The target audience for this research could be Novadust studios themselves to gain more perspective on what their players are experiencing. With more feedback about what is drawing player attention, adjustments can be made to the level design to further avoid players getting lost or confused in the game. Another audience could also be other developers and/or researchers who want to more deeply understand how specific elements in Europa create specific responses or perceptions in players. Using this analysis as a case-study towards some generalizable level design techniques may reveal applications to other future projects.

The results of this analysis will be submitted to Professor Raffaele for evaluation.

Research Questions:

1. What path does the player take through this open game environment?
    a. So we can understand if the critical path is clear and understood for the player.
2. Where does the player spend the most time?
    a. So we can understand what points the player finds interesting.
3. How does the player spend time in each area?
    a. So we can understand what the player enjoys or is trying to do in the game world.

STUDY DESIGN

Observation is an appropriate approach for the aforementioned research questions and intended analysis in order to see what actions players naturally engage with and which directions they choose without solicitation or guidance. Rather than asking participants to self-report on what they see during the experience (which will break their immersion and negatively affect the observation data) or what

they can recall after going through the experience (relying too heavily on memory), it will be more clear and direct to observe players as they are making authentic traversal decisions and executing actions in-game in real time. I want to see if and how the design of game world itself communicates to the player where it wants them to go. In addition to observation data, a brief pre-game questionnaire and semi-structured post-interview may help establish participant background and fill in gaps of interpretation, respectively, that may be unclear if only considering observation data.

Furthermore, if I had more access, I would ideally have two treatment groups that each get a largely similar build of the game with one significant difference either in colors, landmarks, audio cues or other signals to try and determine which ones are actually most effective in communicating different things to the player. However, for this analysis, the participant (and future participants relative to this study) will get the same treatment because I only have access to one build of the game.

Variables to be measured will include:

- RQ1/RQ2 - Time spent at points of interest
- RQ1/RQ2 - Time spent between points of interest,
- RQ3 - What do players do when they reach a point of interest
- RQ3 - What do players do when they are moving between points of interest

*Additional variables to be measured if provided access to debug tools:

- RQ1/RQ2 - Vector map of the path player takes through the game
- RQ1/RQ2 - Number of divergences from the critical path
- RQ3 - Degree of divergences from the critical path

Quantitative time data (time spent), activity data (actions executed) and, if possible, vector data (paths taken via debug tools) can be collected by researcher for each respective research question. Additionally, qualitative survey data will be collected by research moderator via pre-game questionnaire and semi-structured post-interview. The data will be collected either at researcher's home, 370 Jay Street, or another mutually convenient location for the participant and researcher.

Since recruitment efforts are ongoing, a convenient test location has not yet been confirmed.

Instruments will include:

- Learner profile gaming experience pre-questionnaire
- User Observation data/video
    - Observation form
- Semi-structured gaming experience post-interview
- **User Position Logs* (**if provided Debug access**)

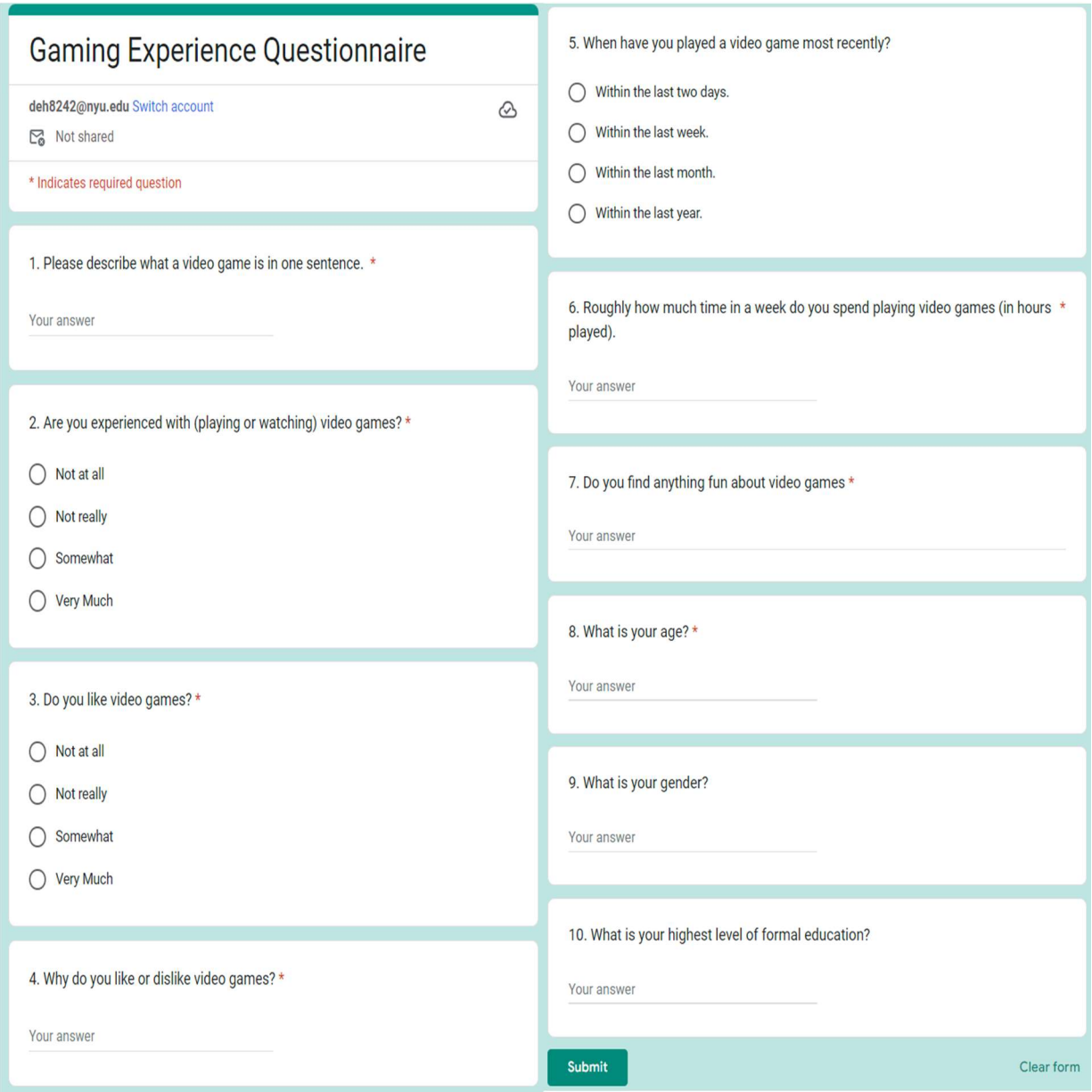
PARTICIPANT

Data will be collected from a group size of one classmate and/or one family member that has little to no experience with playing and/or watching video games. Recruitment efforts are currently still ongoing.

Placeholder Persona

I found this participant through class. She is a graduate level student in the Learning Technology and Experience Design program. She is 27 years old, identifies as an American middle-class woman who works as a dentist's assistant for a living and is not a gamer. She claims not to play video games on any regular basis and hasn't actually picked up a game controller since she was in elementary playing with school friends. She enjoys reggae music, sci-fi books and long walks on the beach in her free time.

INSTRUMENTS & MEASURES



Gaming Experience Questionnaire

deh8242@nyu.edu Switch account

Not shared

* Indicates required question

1. Please describe what a video game is in one sentence. *

Your answer

2. Are you experienced with (playing or watching) video games? *

- Not at all
- Not really
- Somewhat
- Very Much

3. Do you like video games? *

- Not at all
- Not really
- Somewhat
- Very Much

4. Why do you like or dislike video games? *

Your answer

5. When have you played a video game most recently?

- Within the last two days.
- Within the last week.
- Within the last month.
- Within the last year.

6. Roughly how much time in a week do you spend playing video games (in hours played). *

Your answer

7. Do you find anything fun about video games *

Your answer

8. What is your age? *

Your answer

9. What is your gender?

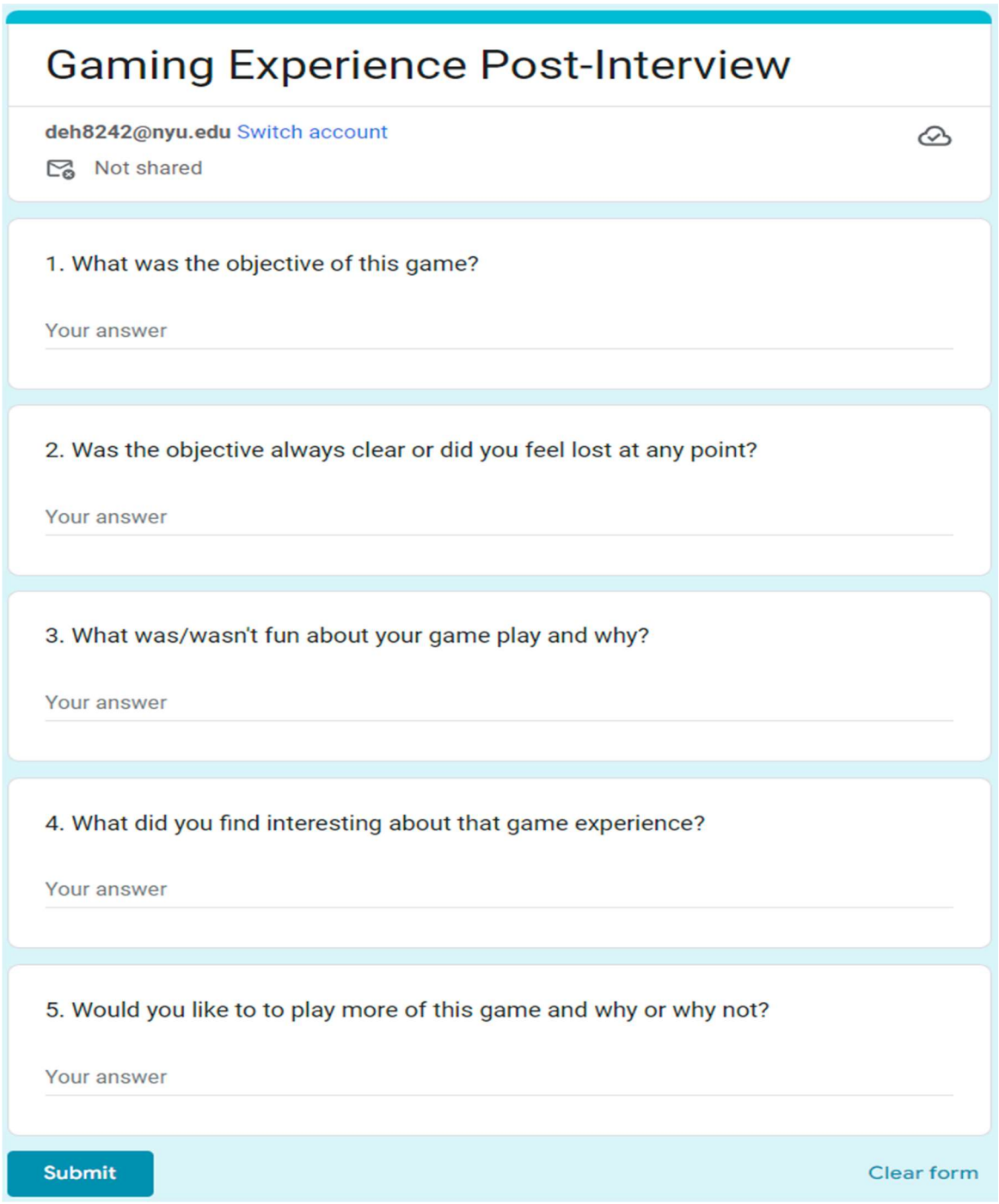
Your answer

10. What is your highest level of formal education?

Your answer

Submit Clear form

Created with Google Forms:
https://forms.gle/5vRAaBraSuebhCpC8



Create with Google Forms:
https://forms.gle/hVBwRfUp2pLnrW8F6

Sample Observation Form

	A	B	C	D	E	F	G	H	I	J
1	**SAMPLE FORM**	POI #1	Toward next POI	POI #2	Toward next POI	POI #3	Toward next POI	POI #4	Toward next POI	POI #5
2	Time Spent									
3										
4										
5										
6	Actions Taken									
7										
8										
9										
10										

PROCEDURE

The participant will sit down at a desk or table and the researcher will then start introductions and explain the focus of the study. The participant will be told that the researchers are trying to assess how fun this experience is. However, researchers are actually trying to understand pathing and pacing and whether or not players can identify and follow the path laid out by game designers or if they are more likely to get lost and lose sight of main objectives in this exploration game. Participants will be given a pre-test. Thereafter a laptop running the game will be provided, the play-test will proceed and lastly, a post-test will be given.

Sample Timetable:

DURATION	TIME	PARTICIPANT	RESEARCHER
-	-:--	Calls Uber	Prepares Test station
0	0:00	Arrives at test location	Greets/Welcomes participant
5m	0:05	Completes Profile Questionnaire	Explains Research Goal
10m	0:15	Completes Play Test	Observes Play Test, Takes notes
5m	0:20	Completes Interview	Conducts Interview
3m	0:23	Calls Uber	Concludes Test and Starts data analysis

The participants will fill out the initial gamer profile questionnaire (5min). The participant will then load up the game and play through the demo while researchers observe and take notes (10min). There will then be a short debriefing interview to ask the participant about their experience (5min). Closing greetings can be exchanged and the participant can be released at which point the researcher can begin data analysis.

DATA ANALYSIS PLAN

The results will be written up into a brief report that includes that participant's profile (results from learner profile questionnaire) and the play test data (time spent at points of interest, time spent between points of interest and actions executed in each respective area). Lastly post-interview response highlights will be summarized and added to the results. If necessary, a brief slide show presentation can additionally be prepared and delivered upon request.

With more participants, the questionnaire and interview data can be collated and aggregated to try and identify user patterns.